

Lab - 2

Basic LAN Setup

1. Peer-to Peer LAN Connection

A Peer-to-Peer (P2P) network consists of two or more computers connected directly without the use of a dedicated server. In a basic P2P setup, two computers are connected using a Copper Cross-Over cable and assigned IP address in the same subnet. After proper configuration, communication is verified using the ping command.

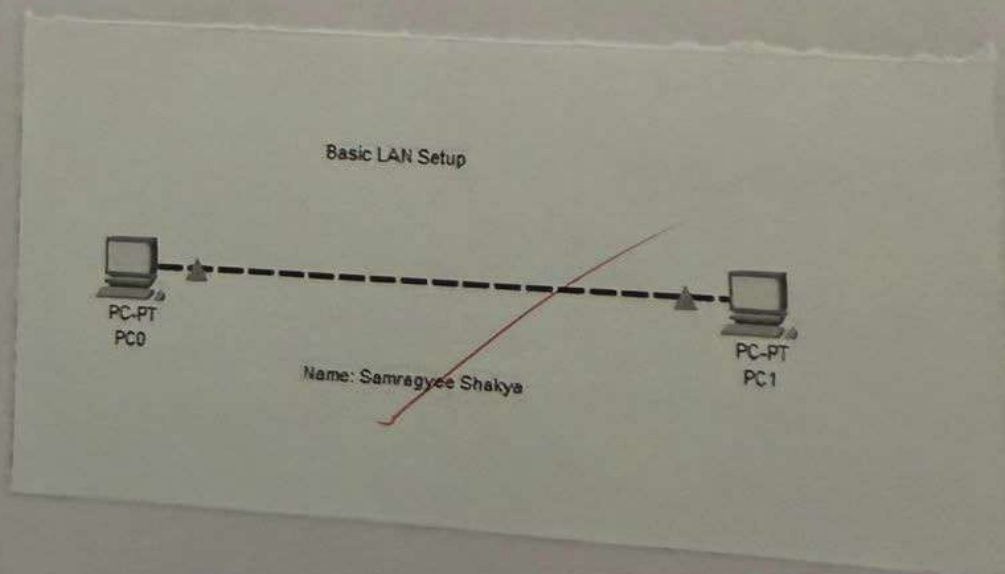
Steps

- In Cisco Packet Tracer, add two PCs (PC0 and PC1).
- Connect PC0 to PC1 using a copper cross-over cable (Fast Ethernet).
- Configure IP address:

PC0 : IP = 192.168.1.1, Subnet Mask = 255.255.255.0

PC1 : IP = 192.168.1.2, Subnet Mask = 255.255.255.0

- Open PC0 → Desktop → Command Prompt and execute the command : ping 192.168.1.2



Result and Discussion:

The two computers were successfully connected using a cross over cable. After assigning valid IP addresses in the same subnet, both devices communicated successfully through the ping command, confirming proper peer-to-peer connectivity. The ping response confirms that Physical Layer, Data Link and Network Layers are functioning properly between two hosts.

Conclusion:

A basic two hosts P2P LAN can be configured using IP address over cross over cable and direct communication between two hosts can be achieved without using any networking device.

2. LAN setup using Hub

A hub is a basic networking device that operates at Physical Layer that connects multiple devices in a LAN and broadcasts every incoming data packets to all connected ports, regardless of the intended destination. Since all devices share the same communication channel, only one device can transmit at a time.

Steps

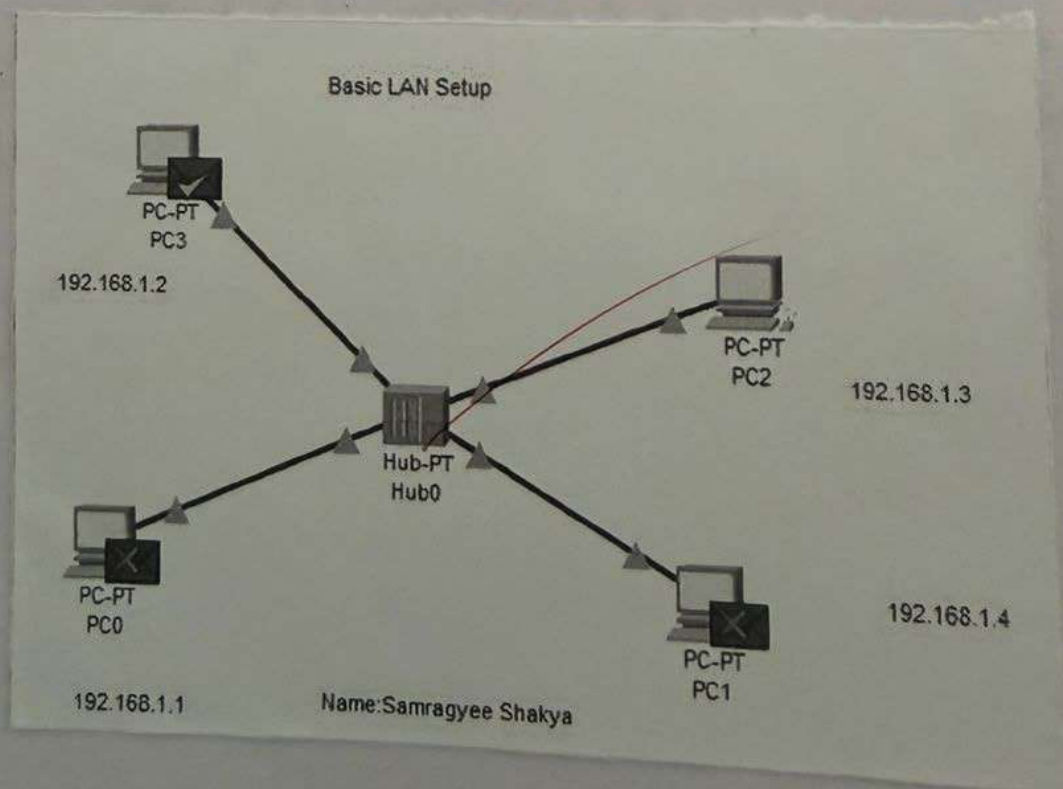
- Add a Hub (Hub0) and 4 PCs (PC0, PC1, PC2 and PC3) and connect them all to Hub using Copper Straight-Through cable.
- Assign IP addresses to all hosts and switch to Simulation Mode.

Testing 1 Packet:

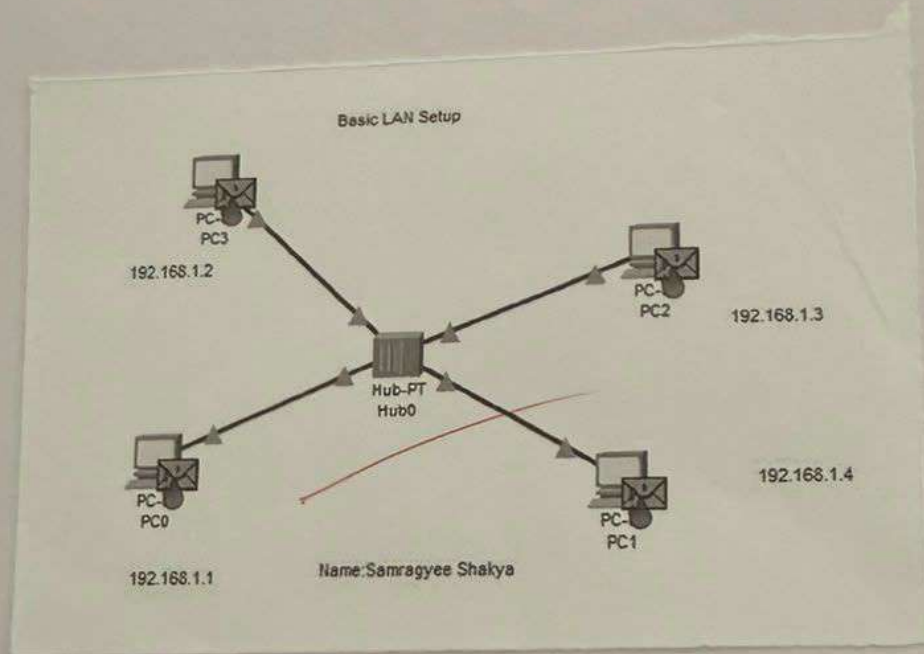
Place a message from PC₃ to PC₂ and simulate.

The envelope (PDU) travels from to Hub0 and then it sends it to PC₀, PC₁ and PC₂ simultaneously.

The intended destination will receive the packet while the others will drop it.



Testing 2 Simultaneous Packets:
 Place a message from PC3 to PC1 and another from PC2 to PC0.
 Simulate it and the both packets go to Hub0 at the same time.
 Since only one device can transmit at a time, a collision occurs
 which is shown by the fire icon appearing on the PCs.



Result and Discussion:

The computers were connected through the hub. When a device sent the message, the hub broadcasted it to every connected port regardless of the destination. All devices received the packet but only the intended device processes it and rest discarded it.

When two devices tried to transmit packet simultaneously, causing a collision and data to be retransmitted, reduced the networking performance.

Conclusion:

The LAN can be setup using hub where it allows communication between devices but broadcasts to all ports, making it less efficient than a switch. However, the setup doesn't allow two or more devices to transmit simultaneously and causes collision.

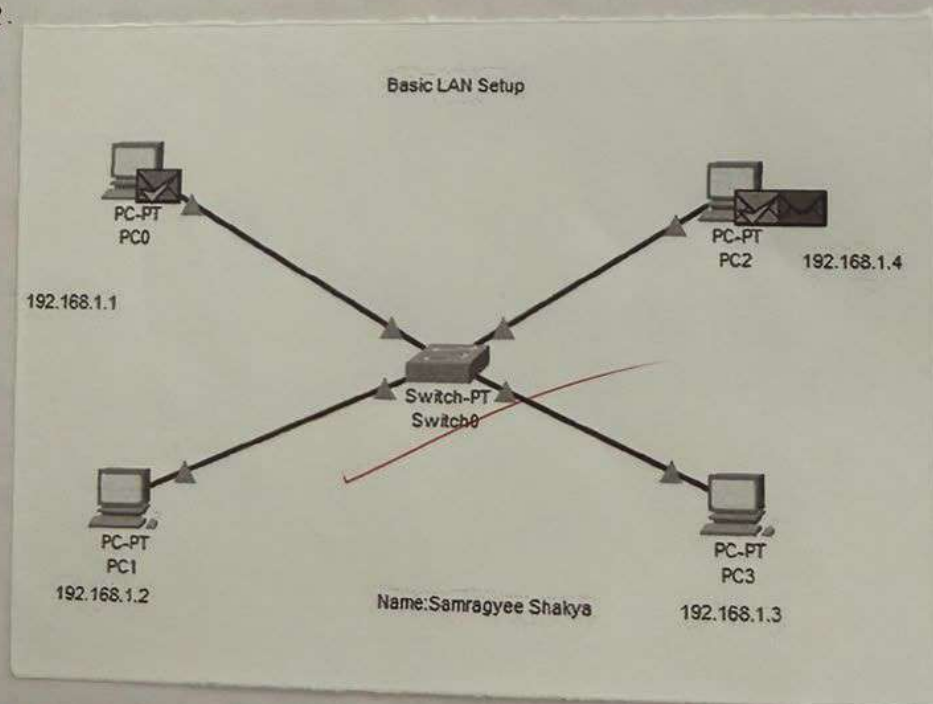


3. LAN Setup using Switch

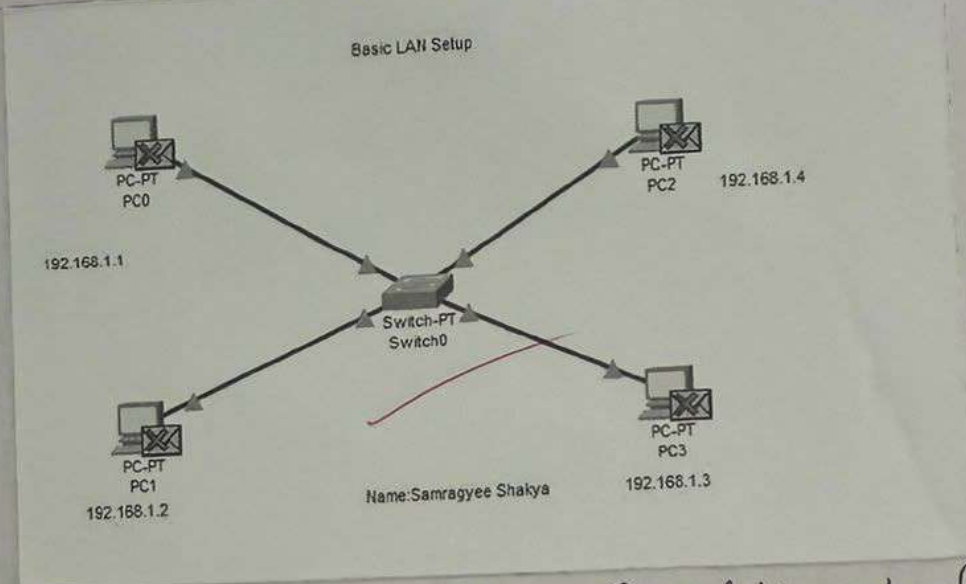
A switch is a networking device that operates at Data Link Layer and connects multiple devices on LAN and forwards data only to the destination device using MAC address table. Each switch ports form a separate collision domain which greatly reduces collision and improves network performance.

Steps

- Add a switch and 4 PCs (PC0, PC1, PC2, PC3) and connect them to switch using Copper Straight Through Cables.
- Assign IP address to all hosts and switch to Simulation mode.



Here, PC0 is the source and PC2 is the destination. The envelope at PC0 represents the ping request being sent. The packet travels from PC0 to switch. The switch checks its MAC address table to determine which port is connected to PC2 and forwards the packet only to that port. Unlike hub, the switch doesn't send the packet to PC1 or PC3, making communication faster and efficient.



Here, the packet has reached PC2. The red X marks don't represent collisions, they simply indicate that the simulation event for those packets has ended or that packet is no longer active.

Result and Discussion:

All computers were connected through switch successfully after assigning IP addresses within same network. The switch forwarded frames from PC0 to PC2 only to the intended destination. Collisions didn't occur in it.

Conclusion:

The LAN network can be setup using Switch making it a Star Topology LAN. The switch can be used for reliable communication among connected hosts without collisions.

Lab 7: Dynamic Routing Implementation using RIP

Theory:

RIP (Routing Information Protocol) is a dynamic routing protocol used by routers to exchange information about the networks they can reach. Unlike static routing, where routes have to be manually entered by the administrator, RIP allows routers to automatically learn and update routes from neighbouring routers. RIP supports max of 15 hops.

The topology consists of three routers. The routers are connected with each other along with switches on Router 0 and Router 2 which is further connected with two PCs.

Steps

1. Add 3 routers, 2 switch and 4 PCs and connect them.
2. Assign IP address to Fast Ethernet / Gigabit interfaces.
3. Verify Connectivity using ping connection.
4. Go to each router, then config and in RIP add the Network Address accordingly.
5. Again ping to verify and trace the route using tracert command.

Discussion:

Using RIP, dynamic routing was done where 3 routers were configured to exchange routing information. Successful ping and traceroute tests confirmed connectivity between the networks.

Conclusion:

Dynamic routing using RIP was successfully implemented. RIP allowed the routers to dynamically learn routes to remote networks.

Lab 8 : VLAN Setup

Theory:

VLAN (Virtual Local Area Network) is a logical division of a physical network into separate broadcast domains. A VLAN allows devices connected to the same physical switch to be separated into different logical networks.

Steps:

1. Add 1 switch and 5 PCs (PC0 to PC4) and connect those PCs with the switch.
2. Go to Config of Switch and add VLAN Database.
3. Assign VLAN to the FastEthernet accordingly.
For PC0 to PC2 : VLAN 100 Student.
For PC3 and PC4 : VLAN 200 Faculty
4. Ping to verify the connection and use show vlan br command to get VLAN names.

Discussion:

VLAN 100 (Student) and VLAN 200 (Faculty) were created and switch ports were assigned to the required VLANs. Connectivity was verified using ping command. The VLANs were used to logically separate connected PCs.

Conclusion:

VLANs provided logical segmentation of the LAN. The configuration was verified using the VLAN database and connectivity testing.

Lab 2: DHCP, DNS and Web Server Setup

Theory:

DHCP (Dynamic Host Configuration Protocol)

DHCP is a protocol used to automatically assign IP address and other Networking Setting like gateway and DNS to devices in network.

DNS (Domain Name System)

DNS is used to convert domain names like www.google.com into IP addresses and helps the user access website using easy memorable name instead of IPs.

Webserver

A webserver hosts and delivers websites to user through HTTP or HTTPS when a browser sends a request, the server responds with appropriate content.

Steps 1:

1. Connect PCs to Switch 0 and server to Switch 1 and make connection between switches too.
2. Go to Server → Desktop → put IP and Default gateway
Server → Services → enable DHCP service.
3. Go to any PCs → Desktop → select DHCP automation.
4. DHCP request will be successful if connection was done.

Step 2:

1. Add Servers to set up and for Server 1, enable DNS and provide IP and Name and address for it.
www.abc.com → 192.168.1.15

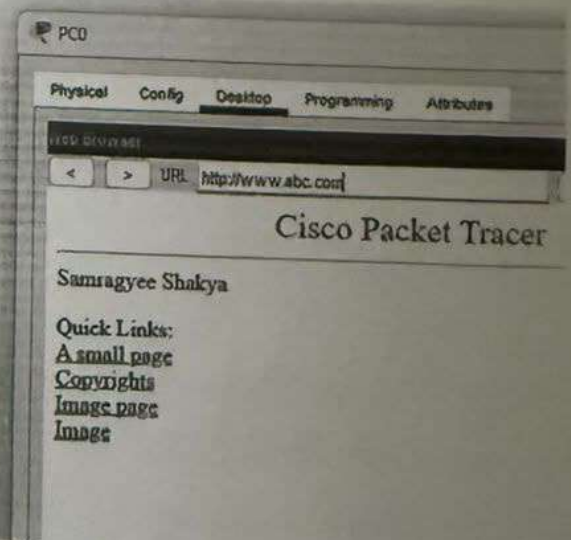
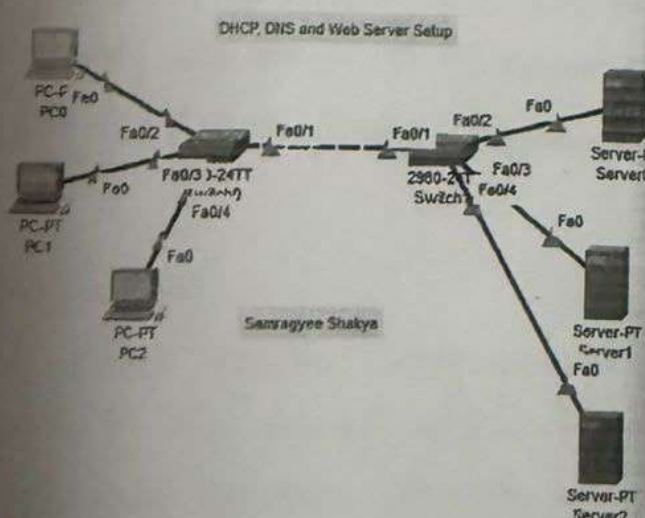
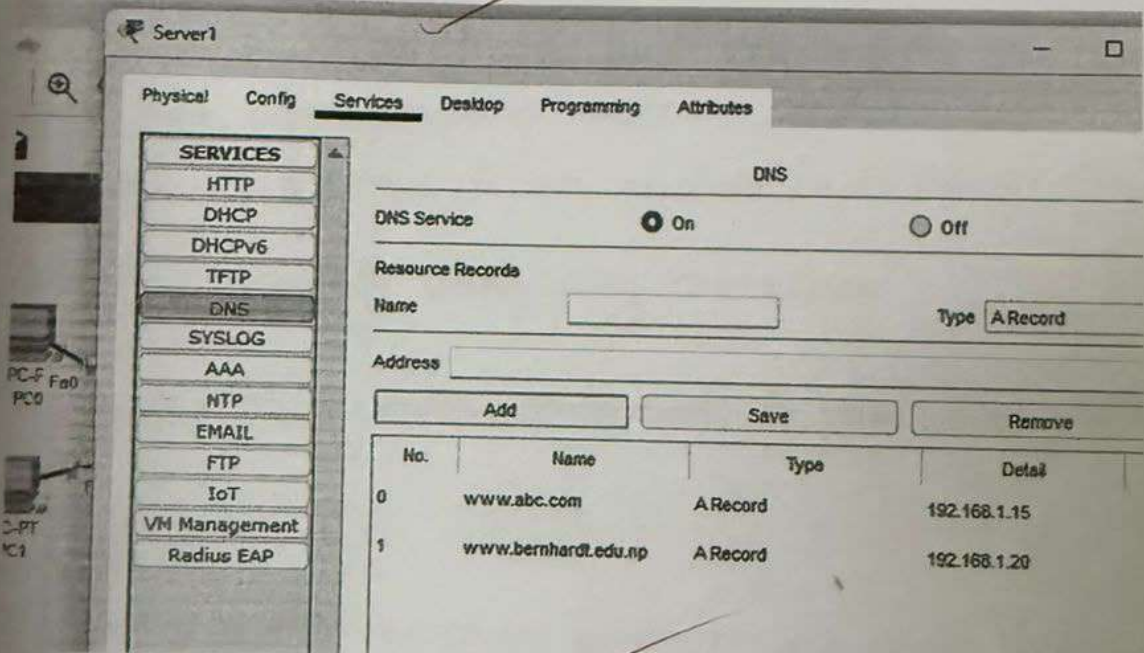
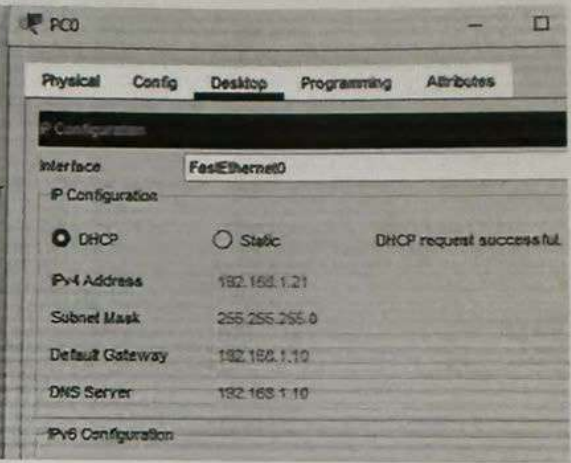
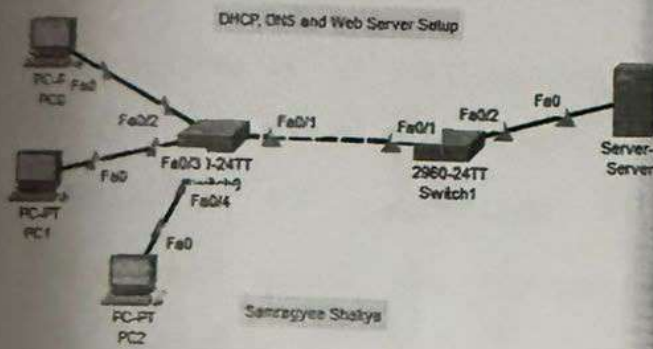
2. For Server 2, add IP configurations and gateways.
Services → HTTP → change in index.html
3. Go to Web browser from any PCs and type the name for it to run.

Discussion:

For DHCP, IPs were automatically assigned and DNS resolved domain names to IPs. Web server delivered the website to the users.

Conclusion:

This lab successfully demonstrated the basic DHCP, DNS and web server setup.



Lab 5 : Basic Router Configuration

Theory:

A router is a networking device that connects different networks and forwards data packets between them based on IP addresses. It uses routing tables to determine the best path for forwarding packets from one network to another.

Steps:

1. Place PCs with switches and connect the two switches with their corresponding routers.
2. Assign IP address to all and to Fast Ethernet / Gigabit interfaces.
3. Verify connectivity using ping connection.

Use router configuration commands to password protect it.

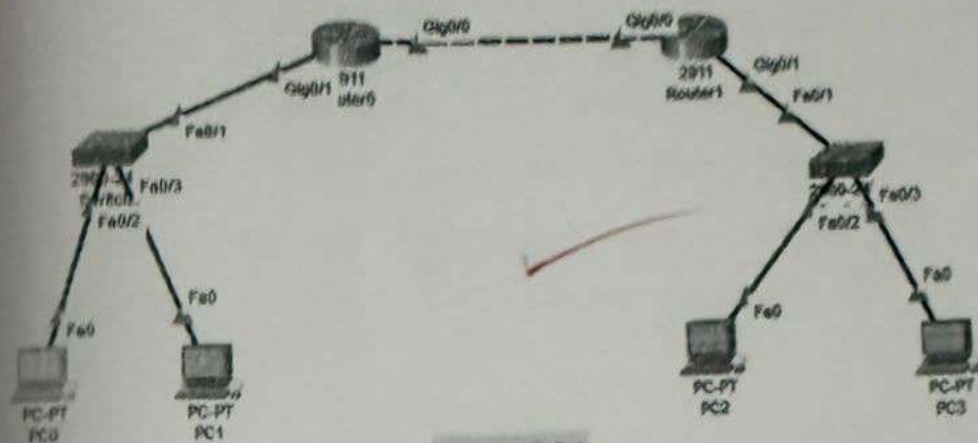
Discussion

IP addresses were assigned to router interfaces, password was configured for security and interfaces were enabled. This experiment showed how routers connect multiple networks. The ping command to PC connected to different router was unsuccessful so Static / dynamic routing is done next.

Conclusion

The basic router configuration was completed. Communication within the local network was established, but not between different routers due to absence of proper routing configuration.

Basic Router Configuration



Samragya Shakya

```

Router(config-if)#exit
Router(config)#enable password 123
Router(config)#exit
Router#
SYS-5-CONFIG_1: Configured from console by console
exit

```

```

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable secret abc
Router(config)#exit
Router#
SYS-5-CONFIG_1: Configured from console by console
Router#exit

```

```

Router#enable
Password:
Router#show running-config
Building configuration...

Current configuration : 704 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
enable secret 5 $1$mERr$Dbe7P1LE0b$nds/DJc0000
enable password 123
!

```

```

ip 0/0
--More--

```

Lab 6: Static Routing Implementation

Theory:

Static routing is a routing technique in which the network administrator manually configures routes in the routing table of a router. Unlike dynamic routing, routers do not automatically exchange routing information. Static routing is simple, secure and suitable for small networks where the network topology rarely changes.

Steps:

1. In the previous Router Configuration, configure IP addresses on all router interfaces, on all PCs.
 2. To configure the static routes,
R0 → Config → Static → Put Network, Mask and Next Hop
R1 → Config → Static → Put Network, Mask and Next Hop
for Next hop, put the ones of opposite router (IP) so that we manually set the path and they can recognize.
3. Perform ping command.

Discussion

Here, Static routing was done between two routers. After which, routers were able to forward packets between different networks. Connectivity was verified using ping command.

Conclusion

The static routing configuration was implemented successfully. Communication between networks was established by manually configuring routes on both routers.

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=5ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 2ms

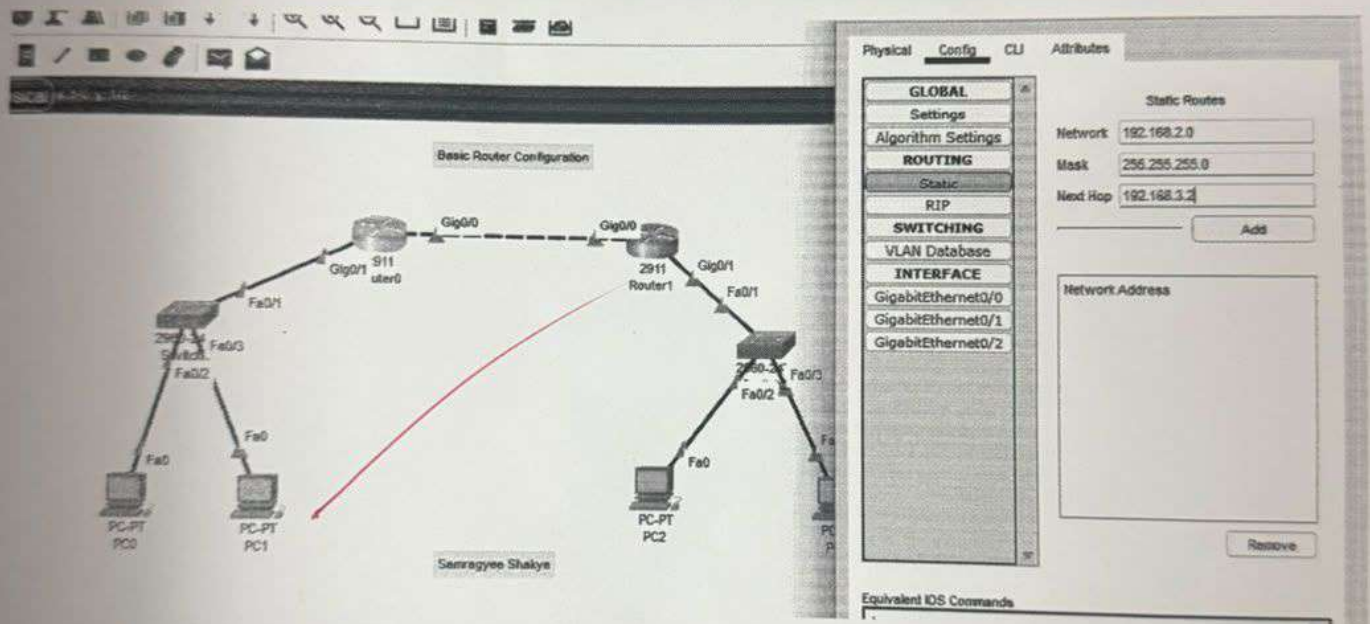
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Request timed out.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```



```

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

```